

Jason Yu

Software Engineer • Tech Speaker

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👤 Work Experience

Boltzbit (Principal Frontend Engineer)

2025 - Present

- Architected a **scalable frontend monorepo** to facilitate seamless integration with external teams.
- Spearheaded the **migration to TypeScript**, improving code reliability and developer experience.
- Led the frontend architecture for a **no-code, AI-empowered app builder**:
 - Developed a **widget registry** enabling users to compose apps via modular components.
 - Created a **Widget SDK** to support third-party extensions and ecosystem growth.

🎯 Fishball Ltd (Founder)

2025 - Present

Building multiple apps, SaaS, and open-source projects:

📱 Mobile Apps

- 📱 **WearDa Metronome**
 - 📱 Jetpack Compose, Kotlin
 - A WearOS vibrating metronome (>12k downloads).
- 🇬🇧 **Ace It: Life in the UK Test**
 - 📱 React Native, TypeScript
 - A study app for the UK Citizenship Test.
- ♠️ **Big2Cal**
 - 📱 React Native, TypeScript
 - Score-tracking app for the popular Hong Kong card game **Big2**.

🌐 Web Apps

- 🀄️ **mjcal.app**
 - 📱 React, TypeScript
 - Mahjong score-tracking app with analytics and stats.

👨🔬 Open Source Projects

- [@fishballpkg/linkz](#)
 - Config-driven static site generator powering my website.
 - Generates a **linktree-like** page with markdown support.
- [@fishballpkg/acme](#)
 - Minimalistic, zero-dependency, pure TypeScript **ACME client**.
 - Uses the **ACME standard** for generating **TLS certificates**.

Bloomberg LP (Senior TypeScript Engineer)

2023 - 2025

- Led migration to TypeScript strict mode
 - Increased visibility of strict mode errors:
 - Developed `@ts-expect-strict-error`, a **TypeScript Language Service Plugin**, showing strict mode errors in IDEs (without affecting `tsc`).
 - Allowed suppressing errors with `@ts-expect-strict-error-next-line`.
 - Prevented new strict mode errors:
 - Built a CLI tool leveraging `@ts-expect-strict-error`.
 - Enforced checks via a new CI step, blocking PRs with unmarked strict errors.
 - Reduced existing strict mode errors:
 - Introduced Grafana dashboards to track progress and a leaderboard to gamify error elimination.
 - Eliminated **15% of 10k errors** in the first month.
- Defined best practices and patterns for scalable TypeScript code.

Attest Technology Limited (Frontend Engineer)

2019 - 2023

- Developed an *A algorithm-based SVG line drawing module**.
- Improved error detection and component testing DX.
- Advocated for the **“rule of least power”** as a coding principle.
- Led migration from **vanilla TypeScript** to **Vue.js**.

The Hut Group (Full Stack Engineer)

2017 - 2019

- Built a **vanilla JavaScript component framework**.
- Founded & led “**JS Club**” to promote modern JavaScript practices.
- Developed a CLI tool to improve developer workflows.

🎤 Public Speaking

- Live Coding: Recreating Vue’s Reactivity System**
 - 📅 Oct 2023
 - 📍 [Vue.js Slovenia Meetup](#)
 - 🎥 [Recording](#)
- Vue 3 Reactivity Reimagined**
 - 📅 Nov 2019
 - 📍 [Vue.js London Meetup](#)
 - 🎥 [Recording](#)
- Live coding: The new Composition API**
 - 📅 Oct 2019
 - 📍 [Vue.js London Conference](#)
 - 🎥 [Recording](#)
- Vue without View - An Introduction to Renderless Components**
 - 📅 Jun 2019
 - 📍 [Vue.js London Meetup](#)
 - 🎥 [Recording](#)
- Building a Simple Virtual DOM from Scratch**
 - 📅 Nov 2018
 - 📍 [Manchester Web Meetup](#)
 - 🎥 [Recording](#)

🔧 Open Source & Side Projects

- [diary.ycmjason.com](#)
 - AI-powered diary using **LLM**, inspired by Tom Riddle’s diary.
 - [Find out more on GitHub](#)
- [Keyboard-Keyboard](#)
 - Web-based **musical keyboard** using **Web Audio API**.
 - [Find out more on GitHub](#)
- [Perfect Colour](#)
 - Vision quiz testing ability to recognize RGB values.
 - [Find out more on GitHub](#)

📝 Articles

📖 All articles availabel at dev.to/ycmjason

Here are some hand-picked highlights:

- Typing Partially Applicable Functions in TypeScript**
 - 📅 Jun 2023
 - 🔗 [Read](#)
- Detecting Barcodes from the Browser**
 - 📅 Nov 2021
 - 🔗 [Read](#)
- Writing Cleaner Code with the Rule of Least Power**
 - 📅 Apr 2020
 - 🔗 [Read](#)
- Building a Simple Virtual DOM from Scratch**
 - 📅 Dec 2018
 - 🔗 [Read](#)

🎓 Education

Imperial College London

2014 - 2017 • BEng Computing

- Courses: Haskell, Java, C, C++, Prolog, Python, AI, Operating Systems, Compilers, Networking.